Identifiability and Predictability Are Distinct: Measured Preimages Across Generative Systems

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Abstract

A recurring proposal in complexity science is to search for the smallest rule that generates an observed structure. We argue that this question is not falsifiable as posed, and replace it with one that is: for a hypothesis space $\mathcal{H}$ of candidate generators and an observation O, how large is the preimage $P(O) = \{h \in \mathcal{H} : O(h) = O(g)\}$, and how does it shrink? We first measure this exactly—by enumeration, not sampling—in three deterministic systems: expressions over $\{1, +, \times\}$, the 256 elementary cellular automata, and a 129-rule family of hypergraph rewriting systems; we then test the resulting framework in a pre-registered fourth system of a deliberately dissimilar kind.

Three results follow. First, the unresolved information decomposes as *observation deficiency* plus *degeneracy of* $\mathcal{H}$; the first term goes to zero in two of our systems and the second is 1.09 bits of irreducible residual in the third. Second, the deficiency term is fixable only where the observer may intervene: given the same automata with the initial state chosen for us rather than by us, a late-arriving observer is left with 2.96 bits and identifies the rule only 20.6% of the time, against 0.00 bits and a guarantee for an observer permitted to set the state. Third, and most consequentially, *identifiability and predictability are distinct, non-redundant properties*: neither determines the other. Elementary cellular automata are identifiable in a single linear pass and yet unpredictable, their generator-to-behaviour map admitting no local structure; hypergraph rewriting under a fixed scheduler is predictable without being identifiable. These examples populate off-diagonal regimes that are not usually examined together within a common finite-preimage framework, separating generator recovery from behavioural prediction.

A pre-registered replication on a fourth and deliberately dissimilar system—all 1,500,625 partially observed 4-state Markov chains under a balanced lumping—meets all five frozen criteria. There the same separation appears for an entirely different reason: the generator is exactly known and the outcome still is not, because the law is stochastic rather than because its behaviour is undecidable. In that system not one generator is uniquely identified by the pre-registered passive observation map at horizon $T = 6$, while 82.0% of them change quadrant once the observer is allowed to set the initial state.

We further find that the forward map is not always a function: identical primitives, rules and iteration counts produce provably non-isomorphic outcomes under different match orders in 2 of 8 rewriting systems tested, so any inverse problem must carry the scheduler as an explicit argument.

1 Introduction

That simple rules generate complex behaviour is not in dispute and has not been for decades. The proposal that recurs, and that motivates this work, is the converse: that one might systematically recover the simple rule beneath a complex object, and that a research programme could be organised around doing so.

We think that programme fails as stated, for a reason that is easy to demonstrate and easy to overlook. Consider the smallest non-trivial generative system available: expressions built from the literal 1 under $+$ and $\times$. The cost of the integer 107—the minimum number of 1s needed to write it—is 16. There is no known closed form for this cost function; whether $\mathrm{cost}(2^k) = 2k$ for all k is open; the cost is not monotone, falling 49 times in $[1, 200]$ as the target increases, with $\mathrm{cost}(107) = 16$ exceeding $\mathrm{cost}(128) = 14$. And the minimum is not unique: 107 admits 26,624 minimal expression trees. If the simplest generative system in existence has no compact, unique, or even stable inverse, "find the simple rule" cannot be the research question.

The question that survives is about *preimages*. Fix a hypothesis space $\mathcal{H}$ of candidate generators and an observation map O. The object of study is

$$P(O) \;=\; \{\, h \in \mathcal{H} \;:\; O(h) = O(g) \,\},$$

the set of generators indistinguishable from the true one under O, and the quantity of interest is $\log_2 |P(O)|$, the bits of generator identity still unresolved. Identifiability is $|P(O)| = 1$.

This framing is not new—it is structural identifiability in the sense of Bellman and Åström [1], a Markov equivalence class in the sense of Verma and Pearl [16], and the Myhill–Nerode construction in the sense of Nerode [12]. Our contribution is not the preimage framing itself, but exact measurements across several generative regimes, followed by a pre-registered test in a deliberately dissimilar stochastic system.

Contributions.

1. A measured decomposition of unresolved information into a term removable by better experiment and a term that is not (§5), with both terms computed exactly rather than estimated.

2. A demonstration that the removable term is removable *only under intervention*, with the passive case measured exactly over every initial condition (§7).

3. Evidence that identifiability and predictability are distinct, non-redundant axes, with all four quadrants populated by measured systems (§10).

4. A pre-registered replication on a fourth system—a family of 1,500,625 partially observed Markov chains—that meets all five frozen criteria (§9), and in which the unpredictability is aleatoric rather than computational.

5. A negative result of practical consequence: the generator-to-behaviour map for elementary cellular automata has no local structure, so search in rule space cannot be guided by proximity (§8).

2 Framework

Let $\mathcal{H}$ be a finite set of candidate generators, G a random element of $\mathcal{H}$ under a uniform prior, and O an observation map. Writing $n_c = |O^{-1}(c)|$ for the size of the class $c \in \operatorname{im} O$, every generator in a class is equiprobable given that class, so the residual uncertainty about generator identity is the conditional entropy

$$H(G \mid O) \;=\; \frac{1}{|\mathcal{H}|} \sum_{c \in \operatorname{im} O} n_c \log_2 n_c \;=\; \mathbb{E}[\log_2 |P(O)|]\,. \tag{1}$$

This is 0 exactly when O separates every generator, and $\log_2 |\mathcal{H}|$ when O is constant. Every figure reported below as "bits unresolved" is this quantity.

We flag one trap explicitly, having fallen into it ourselves. The superficially similar $\log_2 \mathbb{E}[|P(O)|] = \log_2 \big(|\mathcal{H}|^{-1} \sum_c n_c^2\big)$ is *not* equal to (1); by Jensen's inequality it is an upper bound, and the two differ by up to half a bit in our data. It is the wrong quantity here because it is the log of an expected *count*, not an expected *code length*, and does not compose additively the way the decomposition below requires.

Per-generator coordinates. Equation (1) is a single scalar for a whole experiment—an average over the random generator. The claims of this paper are about how quantities *vary between generators*, so they need per-generator coordinates, and we keep the two levels notationally distinct. For an individual $g \in \mathcal{H}$ write

$$i_O(g) \;=\; \log_2 |P_O(g)|, \qquad \text{so that} \qquad H(G \mid O) \;=\; \mathbb{E}_G\big[\, i_O(G)\,\big],$$

the mean unresolved generator information being simply the population average of the per-generator coordinate. Recovery is one question and prediction is another, so alongside it write

$$p_k(g) \;=\; H\big(Y_{t+k} \,\big|\, O_{\le t},\, G = g\big),$$

where Y is a declared predictive target and k a declared horizon. Then i_O is epistemic uncertainty about which mechanism is acting, and p_k is uncertainty about what that mechanism will do *with the mechanism known*: different conditional entropies answering different questions.

Both coordinates are relative to declarations, and the observation horizon is one of them. Everything here is stated with respect to a hypothesis space $\mathcal{H}$, an observation map O—including how much of the process O sees—and an admissible class $\mathcal{A}$; none of the three may be left implicit. In particular, for a stochastic process observed to horizon T, equality of the finite-dimensional law $\mathcal{L}(Y_{0:T-1} \mid g_1) = \mathcal{L}(Y_{0:T-1} \mid g_2)$ does not imply equality of the full process law, so a preimage measured at finite T is an upper bound on the preimage under unlimited passive data and must be reported with T attached.

Every two-axis statement below is about the pair $\big(i_O(g), p_k(g)\big)$, not about the population scalars, and the thesis is the pair of non-implications

$$i_O(g_1) = i_O(g_2) \;\;\not\Longrightarrow\;\; p_k(g_1) = p_k(g_2), \qquad p_k(g_1) = p_k(g_2) \;\;\not\Longrightarrow\;\; i_O(g_1) = i_O(g_2), \tag{2}$$

which is what criteria C2 and C3 of §9 test directly. Where the predictive target is deterministic, as for polynomials, $p_k \equiv 0$ and only i_O has content; §9 works in a setting where both vary.

The decomposition, made exact. Let $\mathcal{A}$ be the family of observations the experimenter is actually permitted to perform—the *admissible class*, which encodes what may be intervened on—and let

$$O^\star \;=\; \operatorname*{arg\,min}_{O' \in \mathcal{A}} H(G \mid O').$$

Define the *intrinsic degeneracy under* $\mathcal{A}$ and the *observation deficiency of* O as

$$D_{\mathcal{H}}(\mathcal{A}) \;=\; H(G \mid O^\star), \qquad \Delta_O \;=\; H(G \mid O) - H(G \mid O^\star).$$

Then trivially, and with both terms non-negative by construction,

$$H(G \mid O) \;=\; \underbrace{\Delta_O}_{\text{a bad experiment}} \;+\; \underbrace{D_{\mathcal{H}}(\mathcal{A})}_{\text{distinct generators, identical objects}}. \tag{3}$$

The content of (3) is not the identity, which is definitional, but the measured values of its two terms and—critically—the fact that $D_{\mathcal{H}}(\mathcal{A})$ moves when $\mathcal{A}$ does. Enlarging the admissible class from passive observation to intervention can convert intrinsic degeneracy into mere deficiency; §7 measures that conversion.

3 Systems and method

All results are exact enumerations over finite hypothesis spaces. Nothing below is sampled unless explicitly stated.

S1: expressions over $\{1, +, \times\}$. Leaves are the literal 1; the budget is the number of leaves. We enumerate every value reachable at each budget and record, for each integer, the minimum budget at which it appears.

S2: elementary cellular automata. A rule is 8 bits—one output per three-cell neighbourhood—so $|\mathcal{H}| = 256$ and there are exactly 8 bits to recover. Rules act on a periodic ring; we use ring size 64 for observation-length curves and 12 (giving 4096 states) where exhaustive state-space analysis is required. A rule is constrained by an observation only where that observation exercises its neighbourhoods; unexercised entries remain free, so the preimage is exactly 2^{8-c} for coverage c.

S3: hypergraph rewriting. States are sets of ordered tuples; a rule rewrites a matched sub-hypergraph. One step applies a maximal non-overlapping set of matches, selected in an order determined by a scheduler seed. Graph comparison uses a Weisfeiler–Leman hash [18], a sound but incomplete invariant: a hash difference proves non-isomorphism, while equality is strong evidence rather than proof. All residual figures for S3 are therefore *lower bounds* on distinguishability.

Validation. Three independent checks against known results. Integer complexity in S1 reproduces OEIS A005245 [13] exactly for $n = 1, \dots, 32$. In S2 the reversible rules are recovered as $\{15, 51, 85, 170, 204, 240\}$, and the 256 rule numbers collapse to 88 equivalence classes under reflection and colour swap. In the Boolean-closure experiment, $\{\wedge, \vee\}$ generates exactly 18 of the 256 functions of three variables, the free distributive lattice on three generators.

Gate set	Functions reached (of 256)	
$\{\wedge\}$	7	
$\{\oplus\}$	8	
$\{\wedge, \vee\}$	18	monotone functions
$\{\oplus, \wedge\}$	128	zero constant term
$\{\oplus, \wedge, 1\}$	256	universal
$\{\wedge, \neg\}$	256	universal
$\{\uparrow\}$	256	universal

Table 1: Boolean closure on three variables. The step from 128 to 256 costs one constant and no new operation.

4 Generation is cheap and already classified

We record these results mainly to set them aside.

Growth class changes discontinuously when a single primitive is added. Over 18 leaves, $\{1, +\}$ reaches exactly n values at budget n; adding $\times$ takes the count to 404 and the per-step growth ratio from 1.06 to 1.37. For polynomials, $\{x, 1, +\}$ reaches only degree ≤ 1 at any budget, while $\{x, 1, +, \times\}$ reaches 12,120 polynomials at budget 10, degrees 0 through 5.

The sharpest transition is Boolean. On three variables, $\{\wedge, \vee\}$ reaches 18 of 256 functions and is stuck there—the monotone functions. $\{\oplus, \wedge\}$ reaches 128, the $\mathbb{F}_2$ polynomials with zero constant term. Adding the single constant 1, with no new operation, reaches all 256: universality crossed by one bit.

None of this is new. Post's lattice [15] classifies every Boolean clone completely, and has since 1941. We report these figures only to establish that the apparatus is calibrated, and to make the point that a programme organised around "which structures does a primitive set generate" is re-entering a field that has been occupied for eighty-five years.

5 The identifiability curve and its plateau

5.1 Cellular automata

Figure-free summary: we compute the mean over all 256 rules of $\log_2 |P(O)|$ as a function of observation length, for five initial conditions on a 64-cell ring.

Initial condition	$T{=}1$	2	3	5	10	25	100	400
single cell	4.00	1.88	1.66	1.60	1.60	1.60	1.58	1.58
density 0.02	4.00	1.88	1.53	1.45	1.45	1.45	1.45	1.45
density 0.10	4.00	1.88	1.53	1.45	1.44	1.44	1.44	1.44
density 0.25	3.00	0.84	0.81	0.81	0.81	0.81	0.81	0.81
density 0.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Table 2: Mean unresolved bits (of 8) against observation length. The curves flatten by $T = 3$; the height at which they flatten is set by the initial condition.

Two readings. A random half-dense row identifies every rule exactly after a *single* step. A single-cell seed leaves 1.58 bits after 400 steps, having left 1.66 after three: running 130 times longer buys 0.08 bits. On this ring, 132 of 256 rules are permanently unidentifiable from a single-cell seed at any observation length. (The count is mildly width-sensitive: a 201-cell ring gives 134, so it should be reported with the ring size attached.)

One well-designed step beats four hundred badly designed ones. This is a demonstration of infomax experimental design in the sense of Lindley [10], not a new principle; what is useful here is that the effect is measured exactly rather than argued.

5.2 Polynomials: the ideal curve and its failure modes

Let $\mathcal{H}$ be integer polynomials of degree $\leq d$ with coefficients in $[-C, C]$, and let O be evaluation at k points. For $d = 4$, $C = 2$ ($|\mathcal{H}| = 3125$, 11.61 bits), consecutive points give $11.61 \to 9.29 \to 5.75 \to 1.06 \to 0$. Degenerate designs never descend: sampling $x = 1$ repeatedly is pinned at 7.90 bits forever, and two points repeated at 5.75.

The threshold arrives earlier than Lagrange interpolation would suggest, and the reason is instructive. Two candidates agree at k points only if their difference—whose coefficients lie in $[-2C, 2C]$—has k roots there, and a small coefficient box cannot supply the large coefficients such a product requires.

Degree	C	$\lvert\mathcal{H}\rvert$	Points to identify	Lagrange bound
3	1	81	3	4
3	2	625	4	4
4	1	243	4	5
4	4	59,049	4	5
6	1	2,187	4	7
6	2	78,125	4	7

Table 3: Shrinking the hypothesis class buys data points. Identifiability is a joint property of $\mathcal{H}$ and O, never of the object alone.

Taking this further, with the design chosen *optimally* rather than merely consecutively, a single point suffices: for $|x| \geq 2C + 1$ evaluation is injective, being positional notation in base x with digits in $[-C, C]$. Thus one sample at $x = 5$ recovers all 11.61 bits, while $x = 1$ leaves 7.90 bits unresolved at any multiplicity. A single well-chosen observation beats infinitely many badly chosen ones, and this is a theorem rather than a tendency.

5.3 Rewriting: a residual that does not vanish

$\mathcal{H}$ is 129 rewrite rules with left-hand side $\{xy\}$ and a right-hand side of one to three edges over x, y, z (7.01 bits). Each ran five steps from a triangle seed under a fixed scheduler seed.

5.4 The decomposition

6 The forward map is not always a function

A formulation of the generative hypothesis we have seen proposed is $C = G(P, R, I)$: complexity as a function of primitives, relations, and iteration. In S3 this is false.

Observation	Classes	Bits unresolved
final $\|E\|$ only	11	4.02
$\|E\|$ trajectory	11	4.02
$\|E\|$ and $\|V\|$ trajectory	13	3.87
final degree sequence	35	2.16
final graph (WL hash)	78	1.09
full graph trajectory	78	1.09

Table 4: Watching the whole trajectory adds nothing over the final state, twice over. The curve bottoms out at 1.09 bits.

System	$\|\mathcal{H}\|$ (bits)	Best observation	Residual
Elementary cellular automata	8.00	0.00	0.00
Polynomials, $d \leq 4$, $C = 2$	11.61	0.00	0.00
Hypergraph rewriting	7.01	1.09	1.09

Table 5: Equation (3) evaluated. For the first two systems every unresolved bit was a bad experiment. For the third, 1.09 bits survive perfect observation: distinct rules generating identical objects.

Applying the same rule to the same seed for the same number of steps, varying only the order in which matches are applied, produced provably non-isomorphic outputs in 2 of 8 rules tested—five distinct outcomes in twelve runs for one rule, three for another. The forward map therefore requires the scheduler as an argument,

$$C = G(P, R, I, S),$$

and whether a given rule is order-independent—causal invariance, in the sense of [20]—is not decidable in general, confluence being undecidable even for restricted rewriting classes [7], though decidable for others. An inverse problem cannot be posed for a forward map that is not a function, so the scheduler must be carried explicitly or held fixed and declared. We hold it fixed and declare it.

We note in passing that growth in S3 is trivially predictable: a right-hand side of k edges gives growth of exactly k per step (1.00, 2.00, 3.00, 4.00 measured). The interesting structure is not in the growth.

7 Without intervention

Every identifiability win above came from *setting the system's state*: choosing a random initial row, choosing evaluation points. These are interventions in the sense of Pearl [14], not more careful passive observation, which is precisely why they were cheap. Equation (3) carries an index for this reason, and this section measures what the index costs.

We take the same 256 automata on a 12-cell ring and forbid the observer to touch anything: nature picks the initial state, and the observer arrives—possibly long after the transient has died—and watches. Computed exactly over all 4096 starts for every rule.

Regime	Bits unresolved	Fully identified
Intervene: choose the state, watch one step	0.00	100%
Passive: nature picks, observer arrives at $t = 0$	0.51	66.9%
Passive: nature picks, observer arrives late	2.96	20.6%

Table 6: The cost of not being allowed to intervene. A state handed to the observer and watched for one step leaves 1.46 bits; a state the observer chose leaves none.

Why watching longer stops helping. A passive observer's entire future is one attractor period; past that the system reshows states it has already shown. On this ring 30.2% of starts land on a fixed point and 42.0% on a period of 1 or 2 (median period 6, longest 240). The cap on identifiability is the *diversity* of the attractor, not the duration of the watch.

The worst case is total. Rules 238–255 leave 7.00 bits: 128 rules permanently indistinguishable. Rule 255 drives every state to all-ones and remains there, so the neighbourhood 111 is the only one that ever occurs and seven of the eight table entries are never tested.

Removing an idealisation. Modelling "nature gives p points" as a uniform random p-subset silently assumes the observation points are spread out, whereas real attractors cluster. We therefore replaced the model with an explicit driving map, so the observation points are whatever its attractor visits.

Model of nature	Mean p	Bits (late)	Identified
Random map on the domain	2.61	1.24	69%
same p, points spread uniformly	2.61	1.23	70%
Affine contraction round$(ax + b)$	1.23	2.56	39%
same p, points spread uniformly	1.23	2.67	40%

Table 7: The clustering idealisation survives: spreading the points out is worth 0.01 bits in one case and -0.11 in the other.

The clustering hypothesis was wrong, and the null result is worth stating plainly: conditioned on p, where the attractor sits barely matters. The damage is in p itself. A dissipative system has a single fixed point, so 77% of the time the late observer sees one value and then that same value forever, and identifiability is settled entirely by where it lands—39% of the time somewhere injective, and otherwise nowhere useful, ever.

Consequence. Passive identifiability is not a smaller number; it is a distribution. With intervention, p is a threshold and past it one always wins. Without, p only sets the odds, and p is set by the system's own dynamics rather than by the observer's patience. In a single-realisation science—one universe, one economic history—the mean is not available either. One gets one draw.

8 The generator-to-behaviour landscape has no local structure

Recovering an elementary cellular automaton rule is trivial: from a random initial condition the exact rule was recovered for all 256 rules in a single linear pass over the spacetime diagram. No search, no minimum-description-length machinery, no symbolic regression.

And the recovered rule licenses no prediction. Taking compressed spacetime size as a computable proxy for output complexity, rule 110 scores 1562 while its eight single-bit neighbours score 145, 146, 151, 308, 329, 440, 783 and 971 — between 0.09× and 0.62×. Across all 2048 single-bit rule mutations the median change is 122 against a full range of 3083, and 10.7% of mutations move complexity by more than a quarter of that range.

The practical consequence is a negative one. A complexity landscape this rough cannot be searched by gradient, hill-climbing, evolutionary, or any other neighbourhood method, because proximity in rule space carries no information about proximity in behaviour. Proposals for an "automated primitive-search engine" that hunts for minimal generators must contend with the fact that, in the richest of our systems, there is no landscape to climb—and separately, that the answer it would return is free anyway.

9 A fourth system, pre-registered

The three systems above are deterministic, fully observed, and algorithmic. If the separation of §10 is an artefact of that setting, a stochastic, partially observed family should break it.

Exploratory run. A first family—all 3375 transition matrices on three hidden states with rows drawn from $\{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$, lumped $\{0, 1\} \to a$, $\{2\} \to b$—was run against three criteria fixed in advance. Two passed; the quadrant-support criterion *failed*, because under passive observation the identifiable row was empty (one generator of 3375). Allowing the observer to reset the hidden state populated all four quadrants, but that was a change of protocol made after seeing the failure, and we report it only as the motivation for what follows.

Confirmatory run. We therefore froze a second design before running it: all $35^4 = 1{,}500{,}625$ transition matrices on *four* hidden states, quarter-grid rows, a balanced $2 + 2$ lumping, horizon $T = 6$, enumerated completely. Passive means a uniform initial state; intervening means the experimenter may initialise each hidden state separately and pool the four resulting laws. Metrics are the per-generator coordinates $i_O(g)$ and $p_1(g)$ of §2, with the quadrant thresholds carried over unchanged ($i_O = 0$; $p_1 < 0.10$ bits/step). Five criteria were fixed: quadrant support at 1% under intervention (C1); a 10–90 percentile spread of p_1 of at least 0.50 bits among identifiable generators (C2); a spread of i_O of at least 1 bit among generators at essentially fixed p_1 (C3); an exact finite-horizon observational-equivalence witness with $p_1 < 0.10$ (C4), meaning generators whose $T = 6$ observable laws coincide exactly; and at least 1% of generators changing quadrant between protocols (C5). C2 and C3 are distributional rather than existence tests, since over 1.5M generators an existence claim is nearly free.

All five criteria were met. C2 gave a spread of 0.55 bits, C3 a spread of 12.00 bits, C4 found 81,482 generators sitting in non-singleton equivalence classes under the exact $T = 6$ observable law while satisfying $p_1 < 0.10$, and C5 recorded 82.0% quadrant migration.

Three things are worth drawing out. First, *not one generator of* 1,500,625 *is uniquely identified by the passive observation map at horizon* $T = 6$—the same empty row that failed

Protocol	id. & pred.	id. & not pred.	not id. & pred.	not id. & not pred.
Passive	0.00%	0.00%	5.43%	94.57%
Intervening	1.85%	80.16%	3.58%	14.41%

Table 8: Quadrant occupancy over all 1,500,625 generators. Passive $H(G \mid O)$ is 3.24 bits over 303,229 distinct laws; intervening, 1.37 bits over 1,235,809.

the exploratory criterion, reproduced on an unrelated family. We have not shown that these generators are structurally non-identifiable under unlimited passive data; the claim is about the declared observation map, and a longer horizon could only shrink the classes. Second, the upper-right quadrant is here populated for a reason entirely unlike the automaton case: the generator is known exactly and the outcome is still uncertain because the law is stochastic, not because its behaviour is undecidable. The same cell is reached by aleatoric and by computational routes, which is evidence that the cell is a real category rather than an artefact of one mechanism. Third, and we think most consequentially, C5 at 82.0% says quadrant membership is largely not a property of the system. The pipeline is

$$(S, \mathcal{A}, \Sigma, O) \longrightarrow (i_O(g),\, p_k(g)) \xrightarrow{\text{thresholds}} Q,$$

so varying $\mathcal{A}$ while holding S fixed moves Q for four generators in five. The consequence for how the taxonomy should be read is direct: *identifiable* and *predictable* are not intrinsic labels attached to a mechanism. They describe a mechanism relative to an observation and intervention regime and a declared prediction target. The intervention result of §7 is the special case of this in which only the number of bits was tracked, not the regime.

The empty passive row should not be read as a discovery. That state aggregation creates observational equivalence classes, and that the inverse lumping problem has no unique solution without further conditions, are established results. Ito, Amari and Kobayashi [6] solve precisely the question of when two different Markov chains generate the same observed process, by linear algebra on the block structure of the transition matrix. What the enumeration adds is the exhaustive quantity—not one generator in 1,500,625, over a completely enumerated class rather than a constructed example—and the contrast with the interventional protocol on the same family. The sharper question, under what conditions on the aggregation the identifiable set is empty rather than merely small, we leave open and do not claim.

One exact witness from the exploratory family states the point most sharply. Five distinct transition matrices induce an identical observable law *at the horizon observed*, with predictive entropy of exactly 0.000 bits per step: 2.32 bits of uncertainty about the cause alongside zero bits of uncertainty about the consequence. Whether the five remain equivalent at every horizon we did not test, and do not claim.

10 Two distinct axes

Identifiability theory grew up in settings where recovering the parameters hands one the trajectory, and treats recovery and prediction as a single achievement.

The claim of this paper rests on §9 alone. There, a single completely enumerated hypothesis class of 1,500,625 generators populates all four combinations of $i_O(g)$ and $p_1(g)$ under one fixed pair of definitions, with criteria frozen before the run—1.85%, 80.16%, 3.58% and 14.41% of the class in the four cells. That is the evidence for (2), and it does not depend on comparing systems to each other.

The three deterministic systems are *not* used to establish the entropy-based quadrant claim, because for them that coordinate is degenerate: with the generator and the current state both known, a deterministic system has $p_k \equiv 0$ by construction, so an elementary cellular automaton is trivially "predictable" in the sense of p_k. What they illustrate instead is that several distinct things get conflated under the word prediction, and that each fails independently of generator recovery.

System	Sense of "prediction"	What generator recovery does not buy
Bounded polynomials	exact future value	nothing: recovery does deliver prediction here. This is the classical case.
Elementary cellular automata	*computational* predictability	the generator is exactly recoverable in one linear pass, while the generator-to-complexity landscape is highly non-smooth (10.7% of single-bit mutations move compressed output size by over a quarter of its full range) and the rule family contains a computationally universal member [4]. Iterating a finite ring is of course always possible; what is not available is any shortcut guided by proximity in rule space.
Hypergraph rewriting	*forward determinacy*	1.09 bits about which rule are unresolved while edge growth is exactly k per step; and with the scheduler free, the outcome is not a function of the rule at all.
Markov family (§9)	$p_1(g)$, entropy of the next observation	the load-bearing case: all four cells occupied under one pair of measures.

Table 9: The deterministic systems illustrate distinct reasons why generator recovery, behavioural computation and forward determinacy should not be conflated. They are not evidence for the entropy-based quadrant claim, which rests on the last row.

We state the resulting claim narrowly, and more narrowly than an earlier draft did. Separating a structural axis from a randomness axis and plotting the pair over an enumerated space of systems is not new: it is the complexity-entropy diagram of computational mechanics [3], discussed in §11. The tempting broad version of our claim—that no existing field has considered identifiability and prediction separately—is an unsupportable negative. So is a claim of statistical independence: across the stochastic family the two axes are weakly *anti*-correlated ($r = -0.30$), which is precisely why we claim distinctness rather than independence. What we demonstrate is this:

> Within one finite-preimage framework and one pair of measures, exact enumeration produces systems occupying all four combinations of the per-generator coordinates $i_O(g)$ and $p_k(g)$; neither coordinate determines the other, in the sense of (2); and the

choice of admissible observation class and of scheduler moves a single system between those regimes.

The off-diagonal cells are the interesting ones, and the reason they are rarely confronted is structural rather than an oversight: the settings where identifiability theory is usually applied are ones in which recovering the parameters does deliver the trajectory, so the question of an exactly known and behaviourally useless generator does not arise there. Since all four cells are occupied inside one hypothesis class under one pair of measures, the thesis needs no cross-system comparison to stand, and we do not rest it on one.

11 Related work

The framing of §2 is structural identifiability [1, 17]; the equivalence-class formulation is standard in causal discovery, where algorithms return a Markov equivalence class rather than a graph [16, 14], and in automata theory via Myhill–Nerode [12]. Optimal experimental design by information gain is due to Lindley [10], with the modern active-learning form in [9]. Generative reachability under a primitive set is universal algebra: Post's lattice [15] classifies the Boolean case completely. Integer complexity is [13] and [8, §F26]. Universality of rule 110 is due to Cook [4]; the broader automaton programme is [19]. Causal invariance in rewriting systems is [20]. The Weisfeiler–Leman invariant is [18], its incompleteness [2]. Algorithmic information theory, which sets the ceiling on compressibility arguments generally, is [11].

The closest prior work, and the comparison a reader should make first, is computational mechanics. Feldman, McTague and Crutchfield [3] plot a measure of structural complexity against a measure of randomness over an enumerated space of processes—maps of the interval, cellular automata, Ising systems, Markov chains—which is the same two-axis move, made eighteen years earlier and over a comparably broad sweep of systems. Our vertical coordinate p_k is, in the stationary limit, their entropy rate h_μ.

The horizontal coordinates differ, and the difference is the whole of what we add. Their statistical complexity C_μ is the memory of the minimal predictor and is a property of the *process*; the ϵ-machine is precisely the canonical representative of an observational equivalence class, and computational mechanics proceeds by quotienting that class away. Our $i_O(g)$ measures the class itself—how many members of a declared hypothesis space $\mathcal{H}$ remain indistinguishable under a declared observation map O—and is therefore a property of g relative to $(\mathcal{H}, O)$ rather than of the process. A process can have small C_μ and large i_O, which is exactly the lower-left quadrant of §10: a simple minimal predictor that many distinct mechanisms share.

That interventions change identifiability is itself well established: active learning of causal models is built on choosing interventions that guarantee it, with optimal strategies known [5]. Our claim is correspondingly narrow. We are not aware of prior work that combines per-generator observational-equivalence size with predictive entropy and exhaustively measures how the resulting joint classification migrates under a change in admissible interventions—$Q = Q(S, \mathcal{A}, \Sigma, O)$, at 82.0% in §9. We make no stronger claim than that.

12 Limitations

Identifiability in §9 is equality of the finite-dimensional law at horizon $T = 6$, not of the full process law. Every preimage reported there is therefore an upper bound on the preimage available to an observer with unlimited passive data, and the empty passive row is a statement about that observation map rather than about structural identifiability in the sense of [6]. The stochastic results of §9 separate cleanly into an exploratory run, whose interventional analysis was post-hoc and is reported as such, and a confirmatory run frozen before execution. Only the latter supports the claim. The quadrant thresholds ($i_O(g) = 0$, $p_1(g) < 0.10$) are conventions carried between the two rather than optimised, but they remain conventions. An earlier version of this manuscript reported $\log_2 \mathbb{E}[|P(O)|]$ for two of the three systems and $H(G \mid O)$ for the third; the figures here are $H(G \mid O)$ throughout, which lowers every affected number (the rewriting residual from 1.55 to 1.09 bits) without changing any qualitative conclusion. The hypothesis spaces are small and finite by design, which is what permits exact enumeration; nothing here is an asymptotic claim. S3 comparisons use a Weisfeiler–Leman invariant, so the 1.09-bit residual is a lower bound on distinguishability and could only grow under a complete test. S3 results hold the scheduler seed fixed, which controls the order-dependence of §6 rather than removing it. The 26,624 figure for minimal expressions of 107 counts ordered trees and so overcounts by commutativity, though not by an order of magnitude. Counts of permanently unidentifiable automata are mildly ring-width sensitive (132 at width 64, 134 at width 201) and should always be reported with the width. The compressed-size proxy for output complexity in §8 is a computable stand-in for an uncomputable quantity and is used only for relative comparison between neighbouring rules.

13 Discussion

The programme this paper argues against—search for the smallest rule generating an observed object—fails in two independent ways. In the simplest system it is ill-posed: minima are non-unique, unstable, and lack a closed form. In the richest it is trivial and unhelpful: the rule is free and predicts nothing.

What replaces it is narrower and testable. Decompose $H(G \mid O)$ by Equation (3), with the admissible class $\mathcal{A}$ stated explicitly rather than left implicit, and report which quadrant of §10 the system occupies. All three are measurable. The first of those questions has now been answered once: §9 carries the classification into stochastic, partially observed dynamics with a hypothesis class four orders of magnitude larger, under criteria fixed in advance. What remains untested is continuous state and continuous time, where exact enumeration is unavailable and the preimage must be estimated rather than counted. We would want that before claiming the quadrant structure is general.

The broader statement the results now support is that a system has no single degree of knowability. What can be known depends on which question is asked—which mechanism is acting, or what it will do next—and on what the observer is permitted to do to it.

Reproducibility. All figures are produced by seven self-contained Python scripts requiring only the standard library: `mgs.py` (algebra and reachability), `mgs_ca.py` (cellular automata), `mgs_rewrite.py` (hypergraph rewriting), `mgs_ident.py` (identifiability curves), `mgs_passive.py` (passive observation), `mgs_stochastic.py` (the exploratory Markov family) and `mgs_preregistered.py` (the frozen confirmatory run, which additionally requires NumPy). All but the last run in a few minutes on one core.

No number in this manuscript is manually trusted. `audit.py` re-derives every quantitative assertion from the script that produces it—recomputing from scratch the few that no committed script emitted—and reports each against the value printed here. It tags provenance as exhaustive computation, sampled computation, derivation, or recomputed in the audit itself, so a reader can see which claims rest on enumeration and which on a fixed seed. The current run is 45 of 45 claims verified, recorded in `AUDIT.txt`, with the values also emitted to `RESULTS.txt` in a form LaTeX can import directly. Bibliographic and interpretive assertions are outside its scope and were checked by hand.

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